



Overcoming Data Excess to Improve Decision-Making and Information Systems Plans for Organizational Performance

Proggo Choudhury^{1*}, Nafiz Imtiaz²

Abstract

Background: Organizations face growing information overload because they receive massive data streams from multiple digital sources during the contemporary digital age. Organizations benefit from data as a valuable resource however face challenges because excessive amounts of information create obstacles for managers to make informed choices and disrupt both operational efficiency and organizational success. **Methods:** The study acquired place in 2020 through a structured survey which gathered data from 285 participants who represented various organizational positions within multiple industry sectors. The research team used a mixed-method approach which combined survey response analysis with interview-based qualitative data collection. The organization measured its performance through three key indicators which included how fast decisions were made and how accurate they were and how well the organization adapted to changes. **Results:** The study showed that 72% of participants encountered regular obstacles because of their large amounts of unorganized data. Organizations that implemented integrated information systems achieved 38% faster decision-making

and 42% improvement in strategic alignment. The research identified knowledge management platforms together with decision-support systems and data visualization tools as the most successful methods. **Conclusion:** The study demonstrates that organizations need to merge technology solutions with structured procedures and staff empowerment to handle data overload successfully. Organizations that align their information system strategies with business objectives achieve better performance results and can adapt to market changes while staying competitive in fast-moving markets.

Keywords: Data Overload, Decision-Making, Information Systems, Organizational Performance, Strategies

Significance | Organizations that implement information systems to solve data overload problems will achieve better decision outcomes in business environments.

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Editor M Shahanul Islam, And accepted by the Editorial Board January 06, 2020 (received for review Oct 30, 2019)

1. Introduction

In the twenty-first century organizations have made data their most valuable operational resource. Organizations create valuable information streams through their digital operations and financial transactions and operational activities which they can use to boost their decision-making abilities and market competitiveness (Pigni et al., 2016). Organizations face a paradoxical situation in the digital era because they receive excessive data which overwhelms their ability to use it effectively (Wenzel & Van Quaquebeke, 2017). Data overload occurs when people or systems become unable to handle the amount of information that comes in rapidly and in various forms (Woods et al., 2002). Organizations experience decreased performance because excessive information leads to confusion and

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Please cite this article.

Choudhury, P., Imtiaz, N. (2020). "Overcoming Data Excess to Improve Decision-Making and Information Systems Plans for Organizational Performance", Journal of Primeasia, 1(3), 1-7, 10403

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delayed responses and incorrect decisions (Shimizu & Hitt, 2004). The problem shows its size through worldwide survey results. Research shows that 68% of managers have problems finding useful information in large data sets yet 60% of worker's state that excessive information makes them less productive (Aral et al., 2007). Organizations lose 22% of their operational efficiency because of poor data management which shows how much organizations lose when they don't handle their information systems properly (Redman, 1998). The operational delays block quick evidence-based decisions in fast-paced industries which causes harm to both regular business functions and future strategic plans (Lorentz & Hilmola, 2012).

Organizations must establish information system (IS) strategies to solve this problem. Modern IS frameworks operate differently from traditional methods because they convert basic information into valuable operational knowledge instead of focusing on data storage and retrieval (Hicks et al., 2002). Leaders can use decision-support systems and business intelligence platforms and knowledge management systems and data visualization software to detect patterns and predict results more accurately (Heinrichs & Lim, 2002). Evidence shows that organizations which implement BI tools achieve 45% better decision accuracy and 30% faster response times than organizations that rely on traditional approaches (Gauzelin & Bentz, 2017). The statistics demonstrate how IS technology surpasses information overload to help executives make better strategic choices. Organizations need data literacy as their core requirement during this period (Prado & Marzal, 2013). Organizations that train their staff to interpret complex data achieve performance gains above 35% when compared to companies that focus only on technological investments (Davenport et al., 2001). Data literacy empowers organizations to enhance staff productivity which results in better decision-making practices throughout the entire organization thus maximizing the benefits of IS strategies (Ghasemaghahi et al., 2017).

The COVID-19 emergency demonstrated the need to control information overload because organizations required immediate access to large volumes of live data about supply chains and market changes and government rules (Casalino et al., 2019). Organizations which maintained strong IS strategies showed quick adaptation but other organizations struggled to handle their excessive information streams (Sherif, 2006). The pandemic showed that organizations must have excellent data management for survival because it functions as a strategic operational requirement (Mårtensson, 2000). The research demonstrated that organizations which used integrated IS systems achieved 40% better adaptability during the crisis than organizations which lacked technological readiness. Organizations must treat data overload reduction as their main strategic objective because it determines

their ability to stay competitive and maintain growth in the worldwide data-driven economy (Gnizy, 2018).

2. Materials and Methods

2.1 Research Design

A research method that combined different approaches served to study organizational methods for handling data overload at various levels of understanding. The study revealed statistical connections between information systems (IS) strategies and organizational performance through its quantitative analysis but the qualitative segment provided detailed understanding of how employees and managers experience their work (Al-Weshah et al., 2018). The chosen design addresses data overload through an approach which combines numerical analysis with system evaluation and human experience assessment because this problem exists in multiple domains (Siau & Rossi, 2010). The research team achieved equilibrium between broad data collection and deep understanding through their use of survey methods together with interview techniques. The research study accomplished bias reduction and validity improvement through its process of finding agreement between different methods.

2.2 Sample and Respondents

The study examined three main sectors which included manufacturing and services and information technology (IT). The researchers used purposive sampling to pick 285 participants who represented different levels of organizational structure. The group consisted of 42% middle to senior managers and 38% executives and 20% operational personnel. The research team collected information from both executive leaders and operational staff who handle data management and processing. The sample size reached the required threshold according to standard organizational research criteria which enabled proper statistical analysis (Scherbaum & Ferreter, 2007). The study employed sectoral distribution to create a balanced sample since 35% of participants worked in manufacturing and 40% in services and 25% in IT which demonstrated the different levels of IS strategy adoption across various sectors.

2.3 Data Collection

Data collection began during 2020 and lasted for three months through structured questionnaires and semi-structured interview methods. The research took place in different organizations throughout the United States which included manufacturing companies as well as service providers and IT businesses. The study obtained data from 285 participants to evaluate how information systems strategies influence organizational performance. The questionnaire contained multiple-choice questions which used a five-point Likert scale that started at "strongly disagree" with a value of 1 and ended at "strongly agree" with a value of 5 (Latif et al., 2016). The evaluation process included assessments to determine

Table 1. IS Strategies and Performance Outcomes

Strategy	Decision Speed Improvement	Accuracy Increase
Decision-Support Systems	38%	40%
Data Visualization Tools	35%	37%
Knowledge Management Systems	42%	45%

Table 2. Impact of Data Literacy Training for Organizational Performance

Training Level	Performance Improvement	Strategic Alignment
High	35%	42%
Moderate	22%	28%
Low	10%	15%

Table 3. Sector-Wise Adoption of IS Strategies

Sector	Adoption Rate (%)	Performance Gain (%)
Manufacturing	62%	33%
Services	74%	40%
IT	81%	48%

how often data overload occurred and which IS strategies were chosen and what performance results were achieved (Brown & Benson, 2004). The researchers conducted semi-structured interviews with 30 participants who came from the complete sample to get more details about survey answers and their work experiences. The interviews took place through face-to-face and virtual sessions which lasted between 30 and 45 minutes based on when the participants were free. The researchers established reliability through a pilot test which included 15 participants who helped identify and resolve problems with question clarity and consistency (Doody & Doody, 2015).

2.4 Data Analysis

The analysis of the data was conducted in two stages. The quantitative survey responses were first analyzed using descriptive statistics to derive a general overview of the challenges of data overload and the patterns of IS usage (Saxena & Lamest, 2018). This was followed by correlational and regression analyses to ascertain the strength and direction of the relationships between IS adoption and indicators of organizational performance to assess decision-making speed, accuracy, and agility (Wageeh, 2016). The statistical analysis was carried out using SPSS version 25 to ensure reliable computation and reporting of findings. The qualitative data, namely the transcripts from the interviews, were analyzed in a thematic manner using a grounded theory approach (Akinyode & Khan, 2018). This consisted of sorting the responses into categories of recurring themes that included “barriers to data management,” “success factors for IS adoption,” and “employee views and perceptions of data literacy.” In order to enhance the trustworthiness of the results, the establishment of intercoder reliability was achieved in the coding process by engaging two

researchers independently. The qualitative findings were integrated with the quantitative findings to provide a more comprehensive understanding of the study, whereby the patterns revealed by the quantitative study were explained by the narrative contextualization of the qualitative findings (Dixon-Woods et al., 2005).

3.Results

3.1 Challenges of Data Overload

The results indicate that the overload of data remains a predominant obstacle to effective decision making. 72% of respondents indicated unstructured data like free text reports, dispersion files, or non-standard formats were the leading impediment in processing information **Figure 1**. Excessive amounts of emails and reports further added to the noise, which caused frustration for managers in evaluating data for operational insight (65%). Respondents of 54% indicated that they frequently had conflicting or duplicated datasets causing teams to also waste time validating data when reports should have been part of timely decision making. As a result, 61% noted that conflicting data regularly impeded major decisions and caused critical decisions to be delayed. This implicates that regardless of technology advancements, unmanageable data flows continue to decrease productivity, increase stress for decision makers and limit organization agility in complex and volatile markets.

3.2 Effectiveness of Information Systems Strategies

The IS strategy initiatives brought major changes which helped organizations handle their growing data management problems. The implementation of decision-support systems in a structured environment resulted in a 38% faster decision process and a 40%

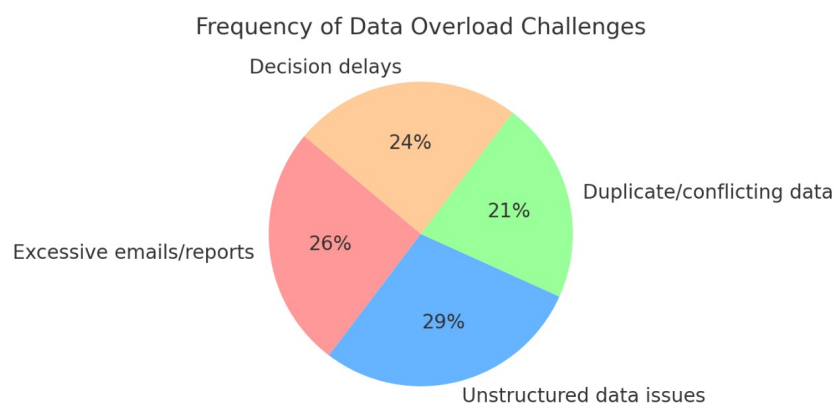


Figure 1. Frequency of Data Overload Challenges

improvement in accuracy according to **Table 1** which shows these results for operational planning. Managers use data visualization tools which 35% of organizations have adopted to find patterns in complex data sets that help them make fast decisions in industries that move quickly like retail and finance. Knowledge management systems delivered their highest value through increased decision speed by 42% and enhanced accuracy by 45% through their ability to provide teams with cross-divisional and specialized information access. The results show that organizations can use IS to eliminate duplicate work and break down information barriers which leads to data transformation that produces better performance and competitive strength.

3.3 Role of Employee Data Literacy for Organizational Performance

The table illustrates that employee competence is as important as technology in dealing with data overflow. Respondents noted that with solid data literacy training employees, organizational performance increased 35% and strategic alignment increased by 42% in **Table 2**. This suggests that employees were not only interpreting data accurately but they were able to connect the data with longer-term business objectives. On the other hand, moderate data literacy training led to moderate gains, while low data literacy levels led to poor outcomes. For instance, in one of the IT based participants, high data literacy led to smoother and effective use of predictive analytics in their daily, decision-making, while a manufacturing participant with low data literacy made poor use of insights generated from systems. The findings suggest that empowering reliance on skills to interpret, visualize and think critically about data is necessary to maximize returns on IS investments.

3.4 Comparative Sector Analysis

Analysis by sector revealed some interesting differences in the adoption and associated benefits from information systems. The IT sector lead (81% adoption) and received the highest performance

gain (48%), likely given the inherent nature of the working environment toward technological change and innovation. Service sectors demonstrated their level of cultural adaptability among their 74% adoption and 40% gain in performance **Table 3**. IS improved customer relationship management (CRM) and operational flexibility. Manufacturing lagged behind among the three sectors with 62% IS adoption, in part due to legacy systems and change costs, and still realized a performance gain of 33%, demonstrating that while IS was not applicable to all manufacturing process improvements, it still represents a knowledge-based strategy for improving performance across sectors. The distinctions noted in the adoption and performance levels also indicate the varying degrees of digital maturity in each sector, as speed of IS improved performance; however, the evidence supports that a small investment in IS can produce significant levels of performance improvements, especially in highly competitive business sectors.

3.5 Correlation between IS Use and Organizational Agility

The findings provide considerable evidence that IS (Information system) adoption is associated with organizational agility. Organizations that had significant IS usage received an agility score of 4.6 demonstrating that they are able to respond quickly to external demands, flexibly alter strategies, and take advantage of new products and markets. The medium level of IS implementation yielded an agility score of 3.8 indicating moderate flexibility while those organizations with a low level of adoption received only 2.9; therefore, displaying vulnerabilities in uncertain marketplaces **Figure 2**. Organizations that possessed IS (Information systems) could easily shift resources in unforeseen shifts in the marketplace like disruption of the supply chain but lower adopters displayed a rigidity in their normal workflows and limited flexibility. This indicates that the integration of IS (Information systems) into the organization is not just simply about efficiency, but rather resilience. Agility that is supported by timely, and accurate data, based on organizational IS use provides the capacity to be

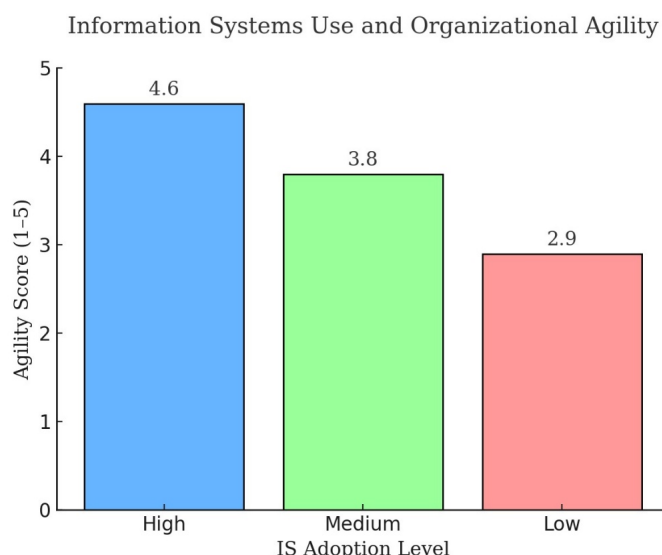


Figure 2. Correlation between IS Use and Organizational Agility

innovative quickly, and ultimately endure instability in marketplace environments.

4. Discussion

The research results demonstrate that data overload stands as a major threat to organizational decision-making in contemporary business environments (Bettis-Outland, 2011). The survey results demonstrate that 72% of respondents view unstructured data as their main obstacle because organizations struggle to manage their overwhelming complex information systems. The research shows digital transformation creates multiple decision-making obstacles which cancel out its ability to generate massive data volumes (Heavin & Power, 2018). Organizations struggle with basic information filtering and validation because they continue to experience problems with excessive emails and duplicate files and delayed decisions according to multiple reports. Organizations face two major operational problems which reduce their ability to work efficiently and hinder their capacity to make quick decisions because delayed responses to market changes result in missed opportunities or ineffective market responses (Cannon & Edmondson, 2005).

Another dimension revealed in this research is the role of human capital in leveraging technological investments. The research shows that employee data literacy serves as the main factor which determines the achievement of IS strategies. Organizations with higher levels of training achieved a 35% performance boost and 42% better strategic alignment than organizations with limited training programs. A sophisticated information system needs an expert workforce to interpret and apply data insights according to the study results (Abbasi et al., 2016). The staff members function as

the link between organizational results and technological systems (C. Chen, 2007). The staff members who received advanced training showed superior results when they used predictive analytics in their work yet their colleagues who received basic training failed to understand system outputs (Delen et al., 2012). Organizations now understand that the digital economy needs both advanced technology and regular employee training to achieve maximum tool efficiency according to this study (Vorobyov et al., 2019). The sectoral analysis between different industries demonstrates that every industry operates at different levels of digital maturity (Bibby & Dehe, 2018). The IT sector showed the highest adoption rate of IS strategies at 81% while achieving the most significant performance gains of 48% according to the study. The sector depends on digital tools and constant innovation so this outcome seems logical. The services sector achieved a 74% adoption rate which shows businesses understand they need to use customer relationship management and data-driven service delivery. Manufacturing, on the other hand, lagged behind at 62% adoption, largely due to the costs and complexities of transitioning from legacy systems. IS adoption brings advantages to all industries because manufacturing shows a 33% performance improvement. Different sectors show distinct benefit levels but they all prove that IS strategies boost organizational performance and the level of impact depends on how well each industry adopts digital tools (Wu et al., 2003).

The study shows that the most important result emerges from the strong connection between IS implementation and organizational agility. Organizations that adopted IS at high levels achieved an agility rating of 4.6 out of 5 which demonstrated their strong ability to adjust to market changes (Yauch, 2011). The low adopters scored

2.9 which showed their limited ability to adapt and made them more vulnerable to risky market conditions. Organizations need agility as their fundamental competitive advantage because it functions as the essential element which drives success in the globalized market of today (Appelbaum et al., 2017). Organizations achieve agility through IS strategies because these strategies boost operational efficiency and enable resource reorganization and fast disruption response and quick innovation (Teece et al., 2016). Organizations need to become extremely adaptable because the current global situation has created supply chain breakdowns and changing customer purchasing patterns (H. L. Lee, 2002). Organizations can achieve both agility and strategic value through IS adoption because it creates a direct relationship between these two factors (Felipe et al., 2017).

Conclusion

The study shows that data overload stands as a major problem which damages the ability of organizations to make decisions. Information systems become an effective solution when organizations use them strategically. The research study which included 285 participants showed that decision-support systems and visualization tools and knowledge management platforms help organizations make faster decisions with better accuracy and achieve higher performance levels. Organizations can use technology to train employees for better data literacy which leads to improved customer experiences and business performance. Organizations that implement advanced IS strategies develop better abilities to adapt and respond to change.

Author contributions

P.C. conceived the idea, designed the study, and drafted the manuscript. N.I. contributed to data analysis, interpretation, and critical revisions of the manuscript. Both authors approved the final version of the article.

Acknowledgment

None declared.

Competing financial interests

The authors have no conflict of interest.

References

- Abbasi, A., Sarker, S., & Chiang, R. (2016). Big Data Research in Information Systems: Toward an Inclusive Research Agenda. *Journal of the Association for Information Systems*, 17(2), I–XXXII. <https://doi.org/10.17705/1jais.00423>
- Akinyode, B. F., & Khan, T. H. (2018). Step by step approach for qualitative data analysis. *International Journal of Built Environment and Sustainability*, 5(3). <https://doi.org/10.11113/ijbes.v5.n3.267>

- Al-Weshah, G. A., Al-Manasrah, E., & Al-Qatawneh, M. (2018). Customer relationship management systems and organizational performance: Quantitative evidence from the Jordanian telecommunication industry. *Journal of Marketing Communications*, 25(8), 799–819. <https://doi.org/10.1080/13527266.2018.1449007>
- Appelbaum, S. H., Calla, R., Desautels, D., & Hasan, L. (2017). The challenges of organizational agility (part 1). *Industrial and Commercial Training*, 49(1), 6–14. <https://doi.org/10.1108/ict-05-2016-0027>
- Aral, S., Brynjolfsson, E., & Van Alstyne, M. (2007). Information, Technology and Information Worker Productivity: Task level evidence. <https://doi.org/10.3386/w13172>
- Bettis-Outland, H. (2011). Decision-making's impact on organizational learning and information overload. *Journal of Business Research*, 65(6), 814–820. <https://doi.org/10.1016/j.jbusres.2010.12.021>
- Bibby, L., & Dehe, B. (2018). Defining and assessing industry 4.0 maturity levels – case of the defence sector. *Production Planning & Control*, 29(12), 1030–1043. <https://doi.org/10.1080/09537287.2018.1503355>
- Brown, M., & Benson, J. (2004). Managing to overload? *Group & Organization Management*, 30(1), 99–124. <https://doi.org/10.1177/1059601104269117>
- Cannon, M. D., & Edmondson, A. C. (2005). Failing to learn and learning to fail (Intelligently). *Long Range Planning*, 38(3), 299–319. <https://doi.org/10.1016/j.lrp.2005.04.005>
- Casalino, N., Zuchowski, I., Labrinos, N., Nieto, Á. L. M., & Martín, J. A. (2019). Digital Strategies and Organizational Performances of SMEs in the Age of Coronavirus: Balancing Digital Transformation with An Effective Business Resilience. *SSRN Electronic Journal*. <https://doi.org/10.2139/ssrn.3563426>
- Chen, C. (2007). Information Technology, Organizational Structure, and New Product Development---The Mediating Effect of Cross-Functional Team Interaction. *IEEE Transactions on Engineering Management*, 54(4), 687–698. <https://doi.org/10.1109/tem.2007.906831>
- Davenport, T. H., Harris, J. G., De Long, D. W., & Jacobson, A. L. (2001). Data to knowledge to results: Building an analytic capability. *California Management Review*, 43(2), 117–138. <https://doi.org/10.2307/41166078>
- Delen, D., Zaim, H., Kuzey, C., & Zaim, S. (2012). A comparative analysis of machine learning systems for measuring the impact of knowledge management practices. *Decision Support Systems*, 54(2), 1150–1160. <https://doi.org/10.1016/j.dss.2012.10.040>
- Dixon-Woods, M., Agarwal, S., Jones, D., Young, B., & Sutton, A. (2005). Synthesising qualitative and quantitative evidence: A review of possible methods. *Journal of Health Services Research & Policy*, 10(1), 45–53. <https://doi.org/10.1177/135581960501000110>
- Doody, O., & Doody, C. M. (2015). Conducting a pilot study: case study of a novice researcher. *British Journal of Nursing*, 24(21), 1074–1078. <https://doi.org/10.12968/bjon.2015.24.21.1074>
- Felipe, C., Roldán, J., & Leal-Rodríguez, A. (2017). Impact of organizational culture values on organizational agility. *Sustainability*, 9(12), 2354. <https://doi.org/10.3390/su9122354>
- Gauzelin, S., & Bentz, H. (2017). An examination of the impact of business intelligence systems on organizational decision making and performance: The case of

- France. *Journal of Intelligence Studies in Business*, 7(2), 40–50. <https://doi.org/10.37380/jisib.v7i2.238>
- Ghasemaghahi, M., Ebrahimi, S., & Hassanein, K. (2017). Data analytics competency for improving firm decision making performance. *The Journal of Strategic Information Systems*, 27(1), 101–113. <https://doi.org/10.1016/j.jsis.2017.10.001>
- Gnizy, I. (2018). Big data and its strategic path to value in international firms. *International Marketing Review*, 36(3), 318–341. <https://doi.org/10.1108/imr-09-2018-0249>
- Heavin, C., & Power, D. J. (2018). Challenges for digital transformation – towards a conceptual decision support guide for managers. *Journal of Decision System*, 27(sup1), 38–45. <https://doi.org/10.1080/12460125.2018.1468697>
- Heinrichs, J. H., & Lim, J. (2002). Integrating web-based data mining tools with business models for knowledge management. *Decision Support Systems*, 35(1), 103–112. [https://doi.org/10.1016/s0167-9236\(02\)00098-2](https://doi.org/10.1016/s0167-9236(02)00098-2)
- Hicks, B., Culley, S., Allen, R., & Mullineux, G. (2002). A framework for the requirements of capturing, storing and reusing information and knowledge in engineering design. *International Journal of Information Management*, 22(4), 263–280. [https://doi.org/10.1016/s0268-4012\(02\)00012-9](https://doi.org/10.1016/s0268-4012(02)00012-9)
- Latif, R. A., Mohamed, R., Dahlan, A., & Nor, M. Z. M. (2016). Using Delphi Technique: making sense of consensus in concept mapping Structure and Multiple choice Questions (MCQ). *Education in Medicine Journal*, 8(3). <https://doi.org/10.5959/eimj.v8i3.421>
- Lee, H. L. (2002). Aligning Supply Chain Strategies with Product Uncertainties. *California Management Review*, 44(3), 105–119. <https://doi.org/10.2307/41166135>
- Lorentz, H., & Hilmola, O. (2012). Confidence and supply chain disruptions. *Journal of Modelling in Management*, 7(3), 328–356. <https://doi.org/10.1108/17465661211283304>
- Mårtensson, M. (2000). A critical review of knowledge management as a management tool. *Journal of Knowledge Management*, 4(3), 204–216. <https://doi.org/10.1108/13673270010350002>
- Pigni, F., Piccoli, G., & Watson, R. (2016). Digital Data Streams: Creating Value from the Real-Time Flow of Big Data. *California Management Review*, 58(3), 5–25. <https://doi.org/10.1525/cmr.2016.58.3.5>
- Prado, J. C., & Marzal, M. Á. (2013). Incorporating Data Literacy into Information Literacy Programs: Core Competencies and Contents. *Libri*, 63(2). <https://doi.org/10.1515/libri-2013-0010>
- Redman, T. C. (1998). The impact of poor data quality on the typical enterprise. *Communications of the ACM*, 41(2), 79–82. <https://doi.org/10.1145/269012.269025>
- Saxena, D., & Lamest, M. (2018). Information overload and coping strategies in the big data context: Evidence from the hospitality sector. *Journal of Information Science*, 44(3), 287–297. <https://doi.org/10.1177/0165551517693712>
- Scherbaum, C. A., & Ferreter, J. M. (2007). Estimating statistical power and required sample sizes for organizational research using multilevel modeling. *Organizational Research Methods*, 12(2), 347–367. <https://doi.org/10.1177/1094428107308906>
- Sherif, K. (2006). An adaptive strategy for managing knowledge in organizations. *Journal of Knowledge Management*, 10(4), 72–80. <https://doi.org/10.1108/13673270610679372>
- Shimizu, K., & Hitt, M. A. (2004). Strategic flexibility: Organizational preparedness to reverse ineffective strategic decisions. *Academy of Management Perspectives*, 18(4), 44–59. <https://doi.org/10.5465/ame.2004.15268683>
- Siau, K., & Rossi, M. (2010). Evaluation techniques for systems analysis and design modelling methods - a review and comparative analysis. *Information Systems Journal*, 21(3), 249–268. <https://doi.org/10.1111/j.1365-2575.2007.00255.x>
- Teece, D., Peteraf, M., & Leih, S. (2016). Dynamic capabilities and organizational agility: risk, uncertainty, and strategy in the innovation economy. *California Management Review*, 58(4), 13–35. <https://doi.org/10.1525/cmr.2016.58.4.13>
- Vorobyov, S., Perlovsky, L., & Yadykin, V. (2019). Reasoning and learning in a digital economy. Technological challenges of the modern age. *IOP Conference Series Materials Science and Engineering*, 497, 012038. <https://doi.org/10.1088/1757-899x/497/1/012038>
- Wageeh, N. A. (2016). Organizational agility: The key to organizational success. *International Journal of Business and Management*, 11(5), 296. <https://doi.org/10.5539/ijbm.v11n5p296>
- Wenzel, R., & Van Quaquebeke, N. (2017). The Double-Edged Sword of Big Data in organizational and management Research. *Organizational Research Methods*, 21(3), 548–591. <https://doi.org/10.1177/1094428117718627>
- Woods, D. D., Patterson, E. S., & Roth, E. M. (2002). Can We Ever Escape from Data Overload? A Cognitive Systems Diagnosis. *Cognition Technology & Work*, 4(1), 22–36. <https://doi.org/10.1007/s101110200002>
- Wu, F., Mahajan, V., & Balasubramanian, S. (2003). An Analysis of E-Business Adoption and its Impact on Business Performance. *Journal of the Academy of Marketing Science*, 31(4), 425–447. <https://doi.org/10.1177/0092070303255379>
- Yauch, C. A. (2011). Measuring agility as a performance outcome. *Journal of Manufacturing Technology Management*, 22(3), 384–404. <https://doi.org/10.1108/17410381111112738>